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United States Patent	12397767
Kind Code	B2
Date of Patent	August 26, 2025
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Disc brake

Abstract

A disk brake includes: a piston, which has a bottomed cylindrical shape, and is configured to press braking members against a braked member; a thrust member configured to thrust the piston through rotation of a rotary member configured to be driven by an electric machine; a rotation stopping member configured to restrict rotation of the thrust member relative to the piston; an engaging portion, which is on an inner side of the piston, and is configured to restrict rotation of the rotation stopping member through engagement; and a fixing member configured to restrict movement of the rotation stopping member in an axial direction relative to the piston.

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Appl. No.:	17/796108
Filed (or PCT Filed):	January 27, 2021
PCT No.:	PCT/JP2021/002742
PCT Pub. No.:	WO2021/157438
PCT Pub. Date:	August 12, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230080487 A1	Mar. 16, 2023

Foreign Application Priority Data

JP	2020-017075	Feb. 04, 2020
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Publication Classification

Int. Cl.: **B60T13/74** (20060101); **F16D65/18** (20060101); F16D121/24 (20120101); F16D125/06 (20120101); F16D125/40 (20120101)

U.S. Cl.:

CPC **B60T13/746** (20130101); **F16D65/18** (20130101); F16D2121/24 (20130101); F16D2125/06 (20130101); F16D2125/40 (20130101)

Field of Classification Search

CPC: F16D (65/18); F16D (2125/40); F16D (2121/24); F16D (2123/00); F16D (2127/06); F16D (65/183); F16D (2066/005); F16D (2065/386); F16D (55/46); F16D (2125/06); F16D (2125/34); F16D (2125/58); F16D (65/14); F16D (65/54); F16D (2127/02); F16D (2127/08); F16D (59/02); F16D (2121/12); F16D (2125/44); F16D (2125/56); F16D (49/08); F16D (55/22); F16D (55/30); F16D (63/006); F16D (65/52); F16D (65/56); B60T (13/741); B60T (13/746); B60T (1/005)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority based on Japanese Patent Application No. 2020-017075 filed on Feb. 4, 2020. All disclosed contents including the Specification, the Scope of the Claims, the Drawings, and the Abstract of Japanese Patent Application No. 2020-017075 are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present invention relates to a disc brake to be used for braking of a vehicle.

BACKGROUND ART

(3) There is known a disc brake including an electric parking brake as a braking device for a vehicle. For example, a disc brake described in JP 5898035 B2 includes an electric actuator and a rotary-to-linear motion conversion mechanism. The rotary-to-linear motion conversion mechanism converts a rotary motion of the electric actuator into a linear motion to thrust a piston. The rotary-to-linear motion conversion mechanism includes a screw member and a linearly movable member. The screw member receives rotation of the electric actuator transmitted to the screw member. The linearly movable member is threadably engaged with the screw member, and is linearly moved by the rotation of the screw member to thrust the piston. The linearly movable member is supported so as to be movable in an axial direction relative to the piston and nonrotatable relative to the piston. More specifically, the linearly movable member has an engagement protruding portion. The engagement protruding portion protrudes radially outward from an outer peripheral surface of the linearly movable member. When the engagement protruding portion of the linearly movable member is engaged with an engagement recessed portion of an inner peripheral surface of the piston, the linearly movable member is allowed to move in the axial direction relative to the piston while being nonrotatable relative to the piston.

SUMMARY OF INVENTION

Technical Problem

(4) In the rotary-to-linear motion conversion mechanism included in the disc brake described in JP 5898035 B2 mentioned above, on an outer peripheral side of the linearly movable member thereof,

specifically, a thrust member that thrusts the piston, the engagement protruding portion that allows the linearly movable member to move in the axial direction while being nonrotatable relative to the piston protrudes radially outward. This thrust member is formed integrally by forging. However, when the engagement protruding portion of the thrust member is formed by forging so as to protrude radially outward, the amount of plastic deformation of a material is large, which causes problems such as an increase in manufacturing cost due to an increase in the number of processing steps. In particular, as the size of the piston increases, the amount of plastic deformation increases, resulting in disadvantageous conditions.

(5) The present invention has an object to provide a disc brake including a rotary-to-linear motion conversion mechanism including a thrust member that is easy to manufacture.

Solution to Problem

(6) According to one embodiment of the present invention, there is provided a disc brake including: a piston, which has a bottomed cylindrical shape, and is configured to press braking members against a braked member; a thrust member configured to thrust the piston through rotation of a rotary member driven by an electric machine; a rotation stopping member configured to restrict rotation of the thrust member relative to the piston; an engaging portion, which is formed on an inner side of the piston, and is configured to restrict rotation of the rotation stopping member and stop the rotation stopping member through engagement; and a fixing portion configured to restrict movement of the rotation stopping member in an axial direction relative to the piston.

(7) The disc brake according to one embodiment of the present invention includes the rotary-to-linear motion conversion mechanism including, in particular, the thrust member that is easy to manufacture.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a sectional view of a main part of a disc brake according to an embodiment of the present invention.

(2) FIG. 2 are views for illustrating an interior of a piston of the disc brake according to the embodiment of the present invention, in which FIG. 2(a) is an end view for illustrating the interior of the piston and FIG. 2(b) is a sectional view taken along the line A-A of FIG. 2(a).

(3) FIG. 3(a) is an end view of the piston included in the disc brake according to the embodiment, and FIG. 3(b) is a sectional view taken along the line B-B of FIG. 3(a).

(4) FIG. 4(a) is an end view of a nut member included in the disc brake according to the embodiment, and FIG. 4(b) is a sectional view taken along the line C-C of FIG. 4(a).

(5) FIG. 5(a) is a front view of a rotation stopping member included in the disc brake according to the embodiment, and FIG. 5(b) is a sectional view taken along the line D-D of FIG. 5(a).

(6) FIG. 6(a) is a front view of a fixing member included in the disc brake according to the embodiment, and FIG. 6(b) is a sectional view taken along the line E-E of FIG. 6(a).

(7) FIGS. 7(a) and 7(b) are views for illustrating an interior of a piston of a disc brake according to another embodiment of the present invention, in which FIG. 7(a) is an end view for illustrating the interior of the piston and FIG. 7(b) is a sectional view taken along the line F-F of FIG. 7(a).

DESCRIPTION OF EMBODIMENTS

(8) Now, a disc brake **1** according to embodiments of the present invention is described with reference to FIG. 1 to FIG. 7. In the following description, for convenience of the description, a right side in FIG. 1 is referred to as one end side of the disc brake **1**, and a left side is referred to as another end side of the disc brake **1**. A right-and-left direction in FIG. 1 corresponds to an axial direction of the disc brake **1**.

(9) As illustrated in FIG. 1, the disc brake **1** according to this embodiment includes a pair of inner

and outer brake pads **2**, **3** and a floating caliper **4**. The pair of inner and outer brake pads **2**, **3** are arranged on both sides of a disc rotor **D** mounted to a rotary portion (not shown) of a vehicle in the axial direction so as to sandwich the disc rotor **D**. The pair of inner and outer brake pads **2**, **3** and the floating caliper **4** are supported by a carrier **5** so as to be movable in the axial direction. The carrier **5** is fixed to a non-rotary portion (not shown) of the vehicle, such as a knuckle. The pair of inner and outer brake pads **2**, **3** correspond to braking members. The disc rotor **D** corresponds to a braked member.

(10) A caliper main body **6**, which is a principal part of the caliper **4**, includes a cylinder portion **7** on a proximal end side (one end side) and a claw portion **8** on a distal end side (another end side). The claw portion **8** extends from the cylinder portion **7** to another end side while passing over the disc rotor **D**. A cylinder **10** is provided inside the cylinder **7**. A piston **25** is fitted into the cylinder **10** so as to be slidable in the axial direction. The cylinder **10** is open on another end side, and is closed on one end side with a bottom portion **13** having a shaft hole **12**. The cylinder **10** has an annular groove **14** in its inner peripheral surface on another end side. A piston seal **16** is provided in the annular groove **14**.

(11) As illustrated in FIG. **1**, also referring to FIG. **2** and FIGS. **3**, the piston **25** has a bottomed cylindrical shape formed of a bottom portion **26** and a cylindrical portion **27**. The piston **25** is arranged so that the bottom portion **26** is opposed to the inner brake pad **2**. The piston **25** is fitted into the cylinder **10** so as to be slidable in the axial direction. As illustrated in FIG. **2** and FIGS. **3**, an outer peripheral surface of the cylindrical portion **27** includes an inclined surface **29** having an annular shape at its one end. The inclined surface **29** has an outer diameter gradually decreasing toward the one end. Meanwhile, the cylindrical portion **27** has on its inner side, a small-diameter opening portion **30** and a large-diameter opening portion **31**. The large-diameter opening portion **31** is continuous from one end of the small-diameter opening portion **30**, and is located on one end side. The large-diameter opening portion **31** has an inner diameter larger than that of the small-diameter opening portion **30**. The small-diameter opening portion **30** is formed so as to have a length in the axial direction significantly larger than that of the large-diameter opening portion **31**. An inner peripheral surface of the large-diameter opening portion **31** and an inner peripheral surface of the small-diameter opening portion **30** are connected to each other through intermediation of a first tapered surface **34** having an annular shape. The first tapered surface **34** has a diameter gradually increasing toward one end side.

(12) The inner peripheral surface of the large-diameter opening portion **31** includes two pairs of flat surface portions **36**, **36**. The flat surface portions **36**, **36** of each pair are opposed to each other. In other words, the flat surface portions **36**, **36** are formed at four positions on the inner peripheral surface of the large-diameter opening portion **31** at 90-degree intervals in a circumferential direction. The flat surface portions **36**, **36** extend over a substantially entire range of the large-diameter opening portion **31** in the axial direction from one end of the large-diameter opening portion **31** to the first tapered surface **34**. In this embodiment, the flat surface portions **36**, **36** are formed at four positions on the inner peripheral surface of the large-diameter opening portion **31**. However, the flat surface portions **36** may be formed at one to three positions, or five or more positions. Each of the flat surface portions **36** corresponds to an engaging portion.

(13) A surface of the bottom portion **26** of the piston **25** on one end side and an inner peripheral surface of the small-diameter opening portion **30** are connected through intermediation of a second tapered surface **39** having an annular shape. The second tapered surface **39** has a diameter gradually increasing toward one end side. Further, an outer peripheral edge portion of the bottom portion **26** of the piston **25** has a plurality of recessed portions **41** arranged at intervals in the circumferential direction. Referring to FIG. **1**, protruding portions **42** on a rear surface of the inner brake pad **2** are engaged with corresponding ones of the recessed portions **41**, respectively. This engagement makes the piston **25** nonrotatable relative to the cylinder **10** and also to the caliper main body **6**. As illustrated in FIG. **1**, an outer peripheral surface of the cylindrical portion **27** of the

piston **25** is in contact with the piston seal **16**. A hydraulic chamber **44** is defined by the piston seal **16** and a sealing member **57** described later between the piston **25** and the bottom portion **13** of the cylinder **10**. In the hydraulic chamber **44**, a hydraulic pressure is supplied from a hydraulic source (not shown) such as a master cylinder or a hydraulic-pressure control unit via a port (not shown) of the cylinder portion **7**. A dust boot **46** is provided between an outer peripheral surface of the cylindrical portion **27** of the piston **25** on another end side and an inner peripheral surface of the cylinder **10** on another end side.

(14) As illustrated in FIG. **1**, a motor gear unit **49** and a rotary-to-linear motion conversion mechanism **50** are provided to the caliper main body **6**. The motor gear unit **49** includes an electric motor **48** being an electric machine and a speed reduction mechanism (not shown). The rotary-to-linear motion conversion mechanism **50** receives rotation of the motor gear unit **49** transmitted to the rotary-to-linear motion conversion mechanism **50**. An electronic control unit **51** that controls drive of the electric motor **48** is connected to the electric motor **48** of the motor gear unit **49**. A parking switch **52** is connected to the electronic control unit **51**. The parking switch **52** is operated to turn a parking brake ON/OFF. The electronic control unit **51** can actuate the parking brake based on a signal from the vehicle regardless of whether or not the parking switch **52** has been operated. The speed reduction mechanism of the motor gear unit **49** includes, for example, a spur gear multi-stage speed reduction mechanism and a planetary gear speed reduction mechanism. An output shaft **55** extending from the motor gear unit **49** is inserted into the shaft hole **12** of the bottom portion **13** of the cylinder **10** so as to be rotatable. The sealing member **57** is provided between an outer peripheral surface of the output shaft **55** and an inner peripheral surface of the shaft hole **12**.

(15) The rotary-to-linear motion conversion mechanism **50** converts a rotary motion of the motor gear unit **49** into a linear motion to thrust the piston **25**, and retains the piston **25** in a braking position. The rotary-to-linear motion conversion mechanism **50** is arranged inside the cylinder **10**. The rotary-to-linear motion conversion mechanism **50** includes a spindle **60** and a nut member **61**. The spindle **60** is a rotary member, and has a male thread portion **63**. The nut member **61** is a thrust member, which is threadably engaged with the spindle **60** to thrust the piston **25**. The spindle **60** has the male thread portion **63** and a flange portion **64**. The flange portion **64** is formed at one end of the male thread portion **63**. The flange portion **64** has an annular shape, and protrudes radially outward from an outer peripheral surface of the male thread portion **63**. The output shaft **55** is connected to one end side of the flange portion **64** of the spindle **60** so as to be nonrotatable relative to the spindle **60**. As a result, a rotational torque output from the motor gear unit **49** is transmitted to the spindle **60**. The male thread portion **63** of the spindle **60** is arranged so that its other end is located in proximity to the bottom portion **26** of the piston **25**. A thrust bearing **65** is provided between one end surface of the flange portion **64** of the spindle **60** and the bottom portion **13** of the cylinder **10**.

(16) As illustrated in FIG. **1**, the nut member **61** is threadably engaged with the male thread portion **63** of the spindle **60**. Also referring to FIGS. **4**, the nut member **61** has a cylindrical shape. Two pairs of flat surface portions **70**, **70** are formed on an outer peripheral surface of the nut member **61**. The flat surface portions **70**, **70** of each pair are opposed to each other. In other words, the flat surface portions **70**, **70** are formed at four positions on the outer peripheral surface of the nut member **61** at 90-degree intervals in the circumferential direction. The flat surface portions **70**, **70** extend over the entire range of the nut member **61** in the axial direction. In this embodiment, the four flat surface portions **70**, **70** are formed at four positions on the outer peripheral surface of the nut member **61**. However, the flat surface portions **70** may be formed at one to three positions, or five or more positions. The nut member **61** has a female thread portion **68** on its inner side. The female thread portion **68** extends from its substantial center in the axial direction to one end side. The nut member **61** has an opening portion **69** on its other end side. The opening portion **69** is continuous with the female thread portion **68**. The opening portion **69** has an inner diameter set larger than that of the female thread portion **68**. The male thread portion **63** of the spindle **60** and

the female thread portion **68** of the nut member **61** are threadably engaged with each other. The nut member **61** corresponds to a thrust member.

(17) A thread engagement portion between the male thread portion **63** of the spindle **60** and the female thread portion **68** of the nut member **61** is set to have a reversed efficiency of 0 or smaller. Thus, the spindle **60** cannot be rotated with thrust acting on the nut member **61** in the axial direction. Specifically, the thread engagement portion can convert the rotational torque output from the spindle **60** into thrust on the nut member **61** in the axial direction, but cannot convert thrust of the nut member **61** in the axial direction into a rotational torque for the spindle **60**. Referring to FIG. 2 and FIGS. 4, an inclined surface **72** is formed on an outer peripheral surface of the nut member **61** at its other end. The inclined surface **72** has an annular shape, and has an outer diameter gradually decreasing toward another end side. An inclination angle of the inclined surface **72** with respect to a radial direction is substantially the same as an inclination angle of the second tapered surface **39** on the inner side of the piston **25** with respect to the radial direction.

(18) As illustrated in FIGS. 2, a rotation stopping member **75** is provided between the inner peripheral surface of the large-diameter opening portion **31** of the piston **25** and the outer peripheral surface of the nut member **61**. The rotation stopping member **75** restricts rotation of the nut member **61** relative to the piston **25**. Also referring to FIGS. 5, the rotation stopping member **75** is formed of a plate member having an annular shape. On an outer peripheral surface of the rotation stopping member **75**, two pairs of short-side portions **77, 77** are formed, and two pairs of long-side portions **78, 78** are formed. The short-side portions **77, 77** of each pair are opposed to each other, and the long-side portions **78, 78** of each pair are opposed to each other. In other words, on the outer peripheral surface of the rotation stopping member **75**, the short-side portions **77** are formed at four positions and the long-side portions **78** are formed at four positions, all of which are arranged alternately in the circumferential direction. In other words, the rotation stopping member **75** has an octagonal shape in plan view. Both of the short-side portion **77** and the long-side portion **78** extend linearly. A length of the long-side portion **78** is set larger than a length of the short-side portion **77**. In this embodiment, on the outer peripheral surface of the rotation stopping member **75**, the short-side portions **77** are formed at four positions and the long-side portions **78** are formed at four positions, all of which extend linearly and are arranged alternately. However, linear portions and arc-shaped portions may be arranged in a mixed manner.

(19) Meanwhile, on an inner peripheral surface of the rotation stopping member **75**, two pairs of linear portions **80, 80** and two pairs of convexly curved portions **81, 81** are formed. The linear portions **80** of each pair are opposed to each other, and the convexly curved portions **81** of each pair are opposed to each other. In other words, on the inner peripheral surface of the rotation stopping member **75**, the linear portions **80** are formed at four positions, and the convexly curved portions **81** are formed at four positions, all of which are arranged alternately in the circumferential direction. The convexly curved portion **81** protrudes in an arc-like manner toward a center of the rotation stopping member **75**. The convexly curved portion **81** may have a linear top portion. The long-side portions **78** of the outer peripheral surface of the rotation stopping member **75** and the convexly curved portions **81** of the inner peripheral surface of the rotation stopping member **75** are opposed to each other, respectively. Meanwhile, the short-side portions **77** of the outer peripheral surface of the rotation stopping member **75** and the linear portions **80** of the inner peripheral surface thereof are opposed to each other.

(20) In this embodiment, on the outer peripheral surface of the rotation stopping member **75**, the short-side portions **77** are formed at four positions so as to correspond to the flat surface portions **36** of the inner peripheral surface of the large-diameter opening portion **31** of the piston **25**. However, the short-side portions **77** are not limited to the arrangement described above. The short-side portions **77** are only required to be arranged so as to correspond to the flat surface portions **36** of the inner peripheral surface of the large-diameter opening portion **31** of the piston **25**. Further, on the inner peripheral surface of the rotation stopping member **75**, the convexly curved portions **81**

are formed at four positions so as to correspond to the flat surface portions **70** of the outer peripheral surface of the nut member **61**. However, the convexly curved portions **81** are not limited to the arrangement described above. The convexly curved portions **81** are only required to be arranged so as to correspond to the flat surface portions **70** of the outer peripheral surface of the nut member **61**.

(21) As illustrated in FIGS. **2**, the rotation stopping member **75** is arranged between the inner peripheral surface of the large-diameter opening portion **31** of the piston **25** and the outer peripheral surface of the nut member **61**. An outer peripheral edge of the rotation stopping member **75** on another end side is brought into abutment against one end of the first tapered surface **34** of the piston **25** at an abutment portion **83**. As is understood from FIG. **2(a)**, the rotation stopping member **75** is arranged so that the short-side portions **77**, **77** of the outer peripheral surface of the rotation stopping member **75** and the flat surface portions **36**, **36** of the inner peripheral surface of the large-diameter opening portion **31** of the piston **25** are in abutment against each other. As a result, the rotation stopping member **75** and the piston **25** are made nonrotatable relative to each other, and are restricted from moving in the radial direction relative to each other. In this embodiment, the rotation stopping member **75** and the piston **25** are made nonrotatable relative to each other by the short-side portions **77** of the outer peripheral surface of the rotation stopping member **75** and the flat surface portions **36** of the inner peripheral surface of the large-diameter opening portion **31** of the piston **25**. However, another relative rotation restriction means such as recess-protrusion engagement, spline engagement, or polygonal engagement may be provided between the outer peripheral surface of the rotation stopping member **75** and the inner peripheral surface of the large-diameter opening portion **31** of the piston **25**.

(22) Further, the rotation stopping member **75** is arranged so that top surfaces of the convexly curved portions **81**, **81** of the inner peripheral surface of the rotation stopping member **75** and the flat surface portions **70**, **70** of the outer peripheral surface of the nut member **61** are brought into abutment against each other. As a result, the rotation stopping member **75** and the nut member **61** are made nonrotatable relative to each other, but the nut member **61** is allowed to move in the axial direction. In this embodiment, the rotation stopping member **75** and the nut member **61** are nonrotatable relative to each other by the convexly curved portions **81** of the inner peripheral surface of the rotation stopping member **75** and the flat surface portions **70** of the outer peripheral surface of the nut member **61**. However, another relative rotation restriction means such as recess-protrusion engagement, spline engagement, or polygonal engagement may be provided between the inner peripheral surface of the rotation stopping member **75** and the outer peripheral surface of the nut member **61**.

(23) As a result, the nut member **61** is made nonrotatable relative to the piston **25** by the rotation stopping member **75**, but is movable in the axial direction. Further, as illustrated in FIG. **2(a)**, sufficient gaps are defined by the outer peripheral surface of the nut member **61** except for the flat surface portions **70** and the linear portions **80** of the inner peripheral surface of the rotation stopping member **75**. Further, sufficient gaps are defined by the inner peripheral surface of the piston **25** except for the flat surface portions **36** and the long-side portions **78** of the outer peripheral surface of the rotation stopping member **75**. These gaps are secured as air communication paths at a time of air ventilation. Thus, residual air, which may remain when a hydraulic pressure is supplied from a hydraulic source to the hydraulic chamber **44** inside the caliper main body **6** (cylinder **10**), can be reduced.

(24) Further, as illustrated in FIGS. **2**, a fixing member **84** is provided on one end side of the rotation stopping member **75**. The fixing member **84** urges the rotation stopping member **75** toward another end side, in other words, toward the abutment portion **83**. The fixing member **84** restricts movement of the rotation stopping member **75** in the axial direction relative to the piston **25**. Also referring to FIGS. **6**, the fixing member **84** is formed of a leaf spring member having an annular shape with a given width. As illustrated in FIG. **2(a)**, an inner diameter of the fixing member **84** is

sufficiently larger than an outer diameter of the nut member **61**. An outer diameter of the fixing member **84** is sufficiently smaller than an inner diameter of the large-diameter opening portion **31** of the piston **25**. The outer diameter of the fixing member **84** and a distance between one pair of long-side portions **78, 78** of the rotation stopping member **75** are substantially the same. Further, as illustrated in FIG. 2 and FIGS. 6, an outer peripheral surface of the fixing member **84** has a plurality of protruding portions **86** that are arranged at intervals in the circumferential direction. The protruding portions **86** protrude radially outward from the outer peripheral surface of the fixing member **84**. Each of the protruding portions **86** has a substantially rectangular shape in plan view. (25) The protruding portions **86** extend while being slightly inclined with respect to a direction orthogonal to an axial direction of the fixing member **84**. Referring to FIG. 2(b), the protruding portions **86** are inclined toward one end side under a state in which the fixing member **84** is placed in the large-diameter opening portion **31** of the piston **25**. An inclination angle α of the protruding portion **86** is appropriately set to fall within a range of from 5 degrees to 30 degrees. In this embodiment, the inclination angle α is set to 15 degrees. In this embodiment, the protruding portions **86** are formed at four positions at 90-degree intervals. A distance between distal ends of one pair of protruding portions **86, 86** opposed to each other is slightly larger than a distance between one pair of flat surface portions **36, 36** of the inner peripheral surface of the large-diameter opening portion **31** of the piston **25**, which are opposed to each other. In this embodiment, two pairs of protruding portions **86, 86**, each of the pairs of protruding portions **86, 86** being opposed to each other, are formed. However, one, three, or four pairs of protruding portions **86** may be formed, and the number of pairs is not limited to the number used in this embodiment.

(26) As illustrated in FIGS. 2, the fixing member **84** is provided inside the large-diameter opening portion **31** of the piston **25** so that the protruding portions **86** are opposed to the flat surface portions **36** of the piston **25**, respectively, and another end surface of the fixing member **84** presses one end surface of the rotation stopping member **75**. When the fixing member **84** is mounted, the protruding portions **86** are elastically deformed (warped) in such a manner that the inclination angle α increases. Thus, the fixing member **84** can easily be inserted into the large-diameter opening portion **31** of the piston **25**. Further, after the fixing member **84** is mounted, the protruding portions **86** of the fixing member **84** remain warped. Thus, the protruding portions **86** are urged so that the distal ends of the protruding portions **86** press the flat surface portions **36** of the inner peripheral surface of the large-diameter opening portion **31** of the piston **25** with restoring forces of the protruding portions **86**.

(27) The fixing member **84** is a leaf spring member, and is sufficiently harder than the piston **25** made of a ferrous material used for forging. Thus, when a force in a removal direction, specifically, a load acting in a direction toward one end side is applied to the fixing member **84**, the distal ends of the protruding portions **86** of the fixing member **84** plastically deform the flat surface portions **36** of the piston **25** in such a manner as to push into the flat surface portions **36** of the piston **25**. As a result, the fixing member **84** is reliably fixed inside the large-diameter opening portion **31** of the piston **25**. Then, the rotation stopping member **75** is pressed (urged) with the restoring forces of the protruding portions **86** of the fixing member **84** against the abutment portion **83** between the outer peripheral edge of the rotation stopping member **75** on another end side and the first tapered surface **34** of the piston **25**, and is retained in this position.

(28) As a result, movement of the rotation stopping member **75** in the axial direction relative to the piston **25** is restricted by the fixing member **84**. Further, as illustrated in FIG. 2(a), sufficient gaps are defined by the outer peripheral surface of the nut member **61** except for the flat surface portions **70** and the inner peripheral surface of the fixing member **84**. Further, sufficient gaps are defined by the outer peripheral surface of the fixing member **84** except for the protruding portions **86** and the inner peripheral surface of the large-diameter opening portion **31** of the piston **25**. These gaps are secured as air communication paths at the time of air ventilation. Thus, residual air, which may remain when a hydraulic pressure is supplied from the hydraulic source to the hydraulic chamber

44 inside the caliper main body 6 (cylinder 10), can be reduced. Further, the rotation stopping member 75 is urged toward the abutment portion 83 with the restoring forces of the protruding portions 86 of the fixing member 84. Thus, backlash in the axial direction between the rotation stopping member 75 and the fixing member 84 and backlash in the radial direction between the rotation stopping member 75 and the piston 25 can be suppressed. As a result, a fitting portion between the fixing member 84 and the rotation stopping member 75 and a fitting portion between the rotation stopping member 75 and the piston 25 are not required to be finished with high accuracy so as to suppress the backlash. Thus, dimensional accuracy can be reduced.

(29) Next, an action of the disc brake 1 according to this embodiment is described.

(30) When a brake pedal (not shown) is stepped on by a driver, a hydraulic pressure generated in accordance with the operation of the brake pedal is supplied to the hydraulic chamber 44 inside the caliper main body 6 (cylinder 10) via a master cylinder and a hydraulic circuit (both not shown). As a result, the piston 25 slides from its original position, which is a position when braking is not performed (see FIG. 1), toward another end side while elastically deforming the piston seal 16, and presses the inner brake pad 2 against the disc rotor D. Subsequently, the caliper 4 is moved by a reaction force of the piston 25 toward one end side relative to the carrier 5, and presses the outer brake pad 3 being in abutment against the claw portion 8 against the disc rotor D. As a result, the disc rotor D is sandwiched between the pair of inner and outer brake pads 2, 3 to generate a frictional force to thereby generate a braking force.

(31) Meanwhile, when the brake pedal is released back by the driver, the supply of the hydraulic pressure from the master cylinder to the hydraulic chamber 44 is stopped to decrease the hydraulic pressure in the hydraulic chamber 44. As a result, a restoring force of the piston seal 16 through its elastic deformation causes the piston 25 to retract to the original position to thereby cancel the braking force. When the amount of movement of the piston 25 increases and exceeds a limit of the piston seal 16 along with wear of the inner and outer brake pads 2, 3, a slip occurs between the piston 25 and the piston seal 16, causing the original position of the piston 25 to shift relative to the caliper main body 6. As a result, even when the inner and outer brake pads 2, 3 wear, a pad clearance is adjusted to a constant amount.

(32) Next, an action of the parking brake to hold a braking state so as to maintain a stop state of the vehicle, which is achieved with the disc brake 1 according to this embodiment, is described.

(33) When the electronic control unit 51 receives an apply command (parking brake actuation command) output as a result of, for example, an operation of the parking switch 52 under a state in which the parking brake is released, the electronic control unit 51 energizes the electric motor 48 of the motor gear unit 49 to rotate the spindle 60 in an applying direction. Then, a rotational torque transmitted to the spindle 60 is transmitted to the nut member 61 via the thread engagement portion between the male thread portion 63 of the spindle 60 and the female thread portion 68 of the nut member 61.

(34) In this case, the nut member 61 is supported by the rotation stopping member 75 so as to be nonrotatable relative to the piston 25 and movable in the axial direction relative to the piston 25. Thus, the nut member 61 moves toward another end side. As a result, the nut member 61 moves together with the piston 25 toward another end side while the inclined surface 72 of the nut member 61 is pressing the second tapered surface 39 of the piston 25 and the nut member 61 is being aligned with the piston 25. Then, when the piston 25 slides toward another end side and presses the inner brake pad 2, the disc rotor D is pressed between the inner and outer brake pads 2, 3 to thereby generate a braking force as described above.

(35) When the pressing force of the nut member 61 on the piston 25, in other words, the braking force reaches a predetermined value set in advance, the electronic control unit 51 stops the energization of the electric motor 48 of the motor gear unit 49 to thereby stop the drive (rotation) of the spindle 60 in the applying direction. Further, the electronic control unit 51 can calculate a pressing force of the nut member 61 on the piston 25 (thrust of the nut member 61) based on, for

example, a current value through the electric motor **48**.

(36) Further, as described above, the thread engagement portion between the male thread portion **63** of the spindle **60** and the female thread portion **68** of the nut member **61** is set to have the reversed efficiency of 0 or smaller. Thus, the rotational torque of the spindle **60** can be converted into thrust of the nut member **61** in the axial direction toward another end side. However, the thrust of the nut member **61** in the axial direction cannot be converted into a rotational torque for the spindle **60**. As a result, when the electronic control unit **51** stops the energization of the electric motor **48**, the stop state can be maintained even with a reaction force of the disc rotor **D** against the pressing force via the piston **25**. As a result, the piston **25** is retained in a braking position, and the actuation of the parking brake is completed.

(37) Meanwhile, when the electronic control unit **51** receives a release command (parking brake release command) output through, for example, an operation of the parking switch **52**, the electronic control unit **51** energizes the electric motor **48** of the motor gear unit **49** to rotate the spindle **60** in a releasing direction. A rotational torque transmitted to the spindle **60** is transmitted to the nut member **61** via the thread engagement portion between the male thread portion **63** of the spindle **60** and the female thread portion **68** of the nut member **61**. Then, when the nut member **61** moves toward one end side, the pressing force of the inner and outer brake pads **2, 3** on the disc rotor **D** is released. When an initial state in which a predetermined distance (clearance) is secured between the inclined surface **72** of the nut member **61** and the second tapered surface **39** of the piston **25** is restored, the electronic control unit **51** stops the energization of the electric motor **48** of the motor gear unit **49**.

(38) The above-mentioned disc brake **1** according to this embodiment includes the nut member **61**, the rotation stopping member **75**, the flat surface portions **36, 36**, and the fixing member **84**. The nut member **61** thrusts the piston **25** through the rotation of the spindle **60** driven by the electric motor **48**. The rotation stopping member **75** restricts the rotation of the nut member **61** relative to the piston **25**. The flat plate portions **36, 36** are formed on the inner side of the piston **25**, and restrict the rotation of the rotation stopping member **75** and stop the rotation stopping member **75** through engagement. The fixing member **84** restricts the movement of the rotation stopping member **75** in the axial direction relative to the piston **25**. This configuration eliminates the need of an engagement protruding portion that protrudes radially outward for the nut member **61**, as otherwise required in the related art. Thus, the number of manufacturing steps (processing steps) for the nut member **61** is decreased, and thus the nut member **61** can easily be manufactured, resulting in a reduction in manufacturing cost of the nut member **61**. Further, an engagement recessed portion, which is required in the related art, is not required for the inner peripheral surface of the piston **25**. A large thickness can be secured for an outer peripheral part of the bottom portion **26** of the piston **25**, and hence ease of forging can be improved. Still further, one kind of nut member **61** is used regardless of a size of the piston **25**. Thus, the number of components can be decreased to reduce work such as control of the components. Further, the fixing member **84** is provided separately from the piston **25**. Thus, the piston **25** is not required to be processed with high accuracy, and hence manufacturing cost of the piston **25** can also be reduced.

(39) Further, in the disc brake **1** according to this embodiment, the rotation stopping member **75** is in the axial direction pressed against and retained at the abutment portion **83** between the outer peripheral edge of the rotation stopping member **75** on another end side and the first tapered surface **34** of the piston **25** by the fixing member **84**. As a result, the movement of the rotation stopping member **75** in the axial direction relative to the piston **25** can be restricted with a simple structure.

(40) Further, in the disc brake **1** according to this embodiment, the rotation of the rotation stopping member **75** is restricted by the flat surface portions **36** of the inner peripheral surface of the large-diameter opening portion **31** of the piston **25**, and a structure of the restriction of the rotation is not complicated. Thus, manufacturing cost can be reduced. Further, the rotation stopping member **75**

restricts the rotation of the nut member **61** relative to the piston **25** and allows the movement of the nut member **61** in the axial direction simply through the engagement with the flat surface portions **36** of the inner peripheral surface of the large-diameter opening portion **31** of the piston **25**. This configuration can also reduce the manufacturing cost because of its simple structure and easy assembly.

(41) Still further, in the disc brake **1** according to this embodiment, the distal ends of the protruding portions **86** of the fixing member **84** plastically deform the flat surface portions **36** in such a manner as to push into the flat surface portion **36** of the inner peripheral surface of the large-diameter opening portion **31** of the piston **25**. The restoring forces of the protruding portions **86** of the fixing member **84** urge the rotation stopping member **75** toward the abutment portion **83** between the outer peripheral edge of the rotation stopping member **75** on another end side and the first tapered surface **34** of the piston **25**. As a result, the fixing member **84** can restrict the movement of the rotation stopping member **75** in the axial direction relative to the piston **25**.

(42) Next, a disc brake **1** according to another embodiment of the present invention is described with reference to FIG. 7. In the description of the disc brake **1** according to another embodiment, only differences from the disc brake **1** illustrated in FIG. 1 to FIG. 6 are described.

(43) The disc brake **1** according to another embodiment does not include the fixing member **84** that restricts the movement of the rotation stopping member **75** in the axial direction relative to the piston **25**, which is included in the disc brake **1** illustrated in FIG. 1 to FIG. 6. In the disc brake **1** according to this embodiment, the rotation stopping member **75** is arranged at a position (abutment portion **83**) between an inner peripheral surface of a large-diameter opening portion **31** of a piston **25** and an outer peripheral surface of a nut member **61**, at which an outer peripheral edge of the rotation stopping member **75** on another end side is in abutment against a first tapered surface **34** of the piston **25**.

(44) After that, when flat surface portions **36** are bent radially inward at one end of the large-diameter opening portion **31** of the piston **25**, bulging portions **89** that protrude inward are formed. In this embodiment, the bulging portions **89** are formed at four positions so as to correspond to the flat surface portions **36**. However, the bulging portions **89** are not limited to such arrangement as long as the bulging portions **89** are formed at two or more positions. An inclined surface **29** is formed on an outer peripheral surface on one end side of the piston **25**. A thickness of the piston **25** at its one end is set to be sufficiently small. Thus, one end of the large-diameter opening portion **31** of the piston **25** can easily be bent inward to form the bulging portions **89**. The bulging portions **89** and the abutment portion **83** can restrict the movement of the rotation stopping member **75** in the axial direction relative to the piston **25**. Each of the bulging portions **89** of the large-diameter opening portion **31** of the piston **25** corresponds to a fixing portion.

(45) In the disc brake **1** according to the embodiment illustrated in FIGS. 7, when the flat surface portions **36** are bent radially inward at one end of the large-diameter opening portion **31** of the piston **25**, a plurality of bulging portions **89** are formed. The bulging portions **89** can restrict the movement of the rotation stopping member **75** in the axial direction relative to the piston **25**. As a result, although one step is additionally required for assembly, total manufacturing cost can be reduced because of the absence of the fixing member **84** (see FIG. 1).

(46) As the above-mentioned disc brake **1** according to this embodiment, for example, the following aspects are conceivable.

(47) According to a first aspect, there is provided a disc brake including: a piston (**25**), which has a bottomed cylindrical shape, and is configured to press braking members (**2, 3**) against a braked member (**D**); a thrust member (**61**) configured to thrust the piston (**25**) through rotation of a rotary member (**60**) driven by an electric machine (**48**); a rotation stopping member (**75**) configured to restrict rotation of the thrust member (**61**) relative to the piston (**25**); an engaging portion (**36**), which is formed on an inner side of the piston (**25**), and is configured to restrict rotation of the rotation stopping member (**75**) and stop the rotation stopping member (**75**) through engagement;

and a fixing portion (84, 89) configured to restrict movement of the rotation stopping member (75) in an axial direction relative to the piston (25).

(48) According to a second aspect, in the first aspect, the fixing portion (84, 89) is a fixing member (84) separate from the piston (25).

(49) According to a third aspect, in the second aspect, the rotation stopping member (75) is in the axial direction pressed against and retained at an abutment portion (83) between the rotation stopping member (75) and the piston (25) by the fixing member (84).

(50) According to a fourth aspect, in the first aspect, the fixing portion (84, 89) is a portion (89) formed by inwardly bending an inner peripheral surface of an opening portion (31) of the piston (25) in a radial direction of the piston (25).

(51) According to a fifth aspect, in any one of the first to fourth aspects, the engaging portion (36) is a flat surface portion (36) being at least a part of an inner peripheral surface of an opening portion (31) of the piston (25).

(52) According to a sixth aspect, in the fifth aspect, the rotation stopping member (75) is engaged with the flat surface portion (36) to restrict rotation of the thrust member (61) and allow the thrust member (61) to move in the axial direction.

(53) According to a seventh aspect, in the fifth or sixth aspect, the fixing portion (84) has a protruding portion (86) extending in a radial direction of the piston (25), and the protruding portion (86) plastically deforms the flat surface portion (36) to restrict the movement of the rotation stopping member (75) in the axial direction.

(54) Note that, the present invention is not limited to the embodiments described above, and includes further various modification examples. For example, in the embodiments described above, the configurations are described in detail in order to clearly describe the present invention, but the present invention is not necessarily limited to an embodiment that includes all the configurations that have been described. Further, a part of the configuration of a given embodiment can replace the configuration of another embodiment, and the configuration of another embodiment can also be added to the configuration of a given embodiment. Further, another configuration can be added to, deleted from, or replace a part of the configuration of each of the embodiments.

REFERENCE SIGNS LIST

(55) 1 disc brake, 2 inner brake pad (braking member), 3 outer brake pad (braking member), 25 piston, 31 large-diameter opening portion, 36 flat surface portion (engaging portion), 48 electric motor (electric machine), 50 rotary-to-linear motion conversion mechanism, 60 spindle (rotary member), 61 nut member (thrust member), 75 rotation stopping member, 83 abutment portion, 84 fixing member, 86 protruding portion, 89 bulging portion (fixing portion), D disc rotor (braked member)

Claims

1. A disc brake, comprising: a piston, which has a bottomed cylindrical shape, and is configured to press braking members against a braked member; a thrust member configured to thrust the piston through rotation of a rotary member configured to be driven by an electric machine; a rotation stopping member configured to restrict rotation of the thrust member relative to the piston; an engaging portion, which is on an inner side of the piston, and is configured to restrict rotation of the rotation stopping member through engagement; and a fixing member, which is separate from the piston and the rotation stopping member, and is configured to restrict movement of the rotation stopping member in an axial direction relative to the piston.

2. The disc brake according to claim 1, wherein the rotation stopping member is configured to be pressed in the axial direction against and retained at an abutment portion between the rotation stopping member and the piston by the fixing member.

3. The disc brake according to claim 2, wherein the engaging portion is a flat surface portion

defining at least a part of an inner peripheral surface of an opening portion of the piston.

4. The disc brake according to claim 1, wherein the engaging portion is a flat surface portion defining at least a part of an inner peripheral surface of an opening portion of the piston.

5. The disc brake according to claim 4, wherein the fixing member has a protruding portion extending in a radial direction of the piston, and wherein the protruding portion is configured to plastically deform the flat surface portion to restrict the movement of the rotation stopping member in the axial direction.

6. The disc brake according to claim 4, wherein the rotation stopping member is configured to engage with the flat surface portion to restrict rotation of the thrust member and allow the thrust member to move in the axial direction.

7. The disc brake according to claim 6, wherein the fixing member has a protruding portion extending in a radial direction of the piston, and wherein the protruding portion is configured to plastically deform the flat surface portion to restrict the movement of the rotation stopping member in the axial direction.

8. The disc brake according to claim 3, wherein the rotation stopping member is configured to engage with the flat surface portion to restrict rotation of the thrust member and allow the thrust member to move in the axial direction.

9. The disc brake according to claim 8, wherein the fixing member has a protruding portion extending in a radial direction of the piston, and wherein the protruding portion is configured to plastically deform the flat surface portion to restrict the movement of the rotation stopping member in the axial direction.
